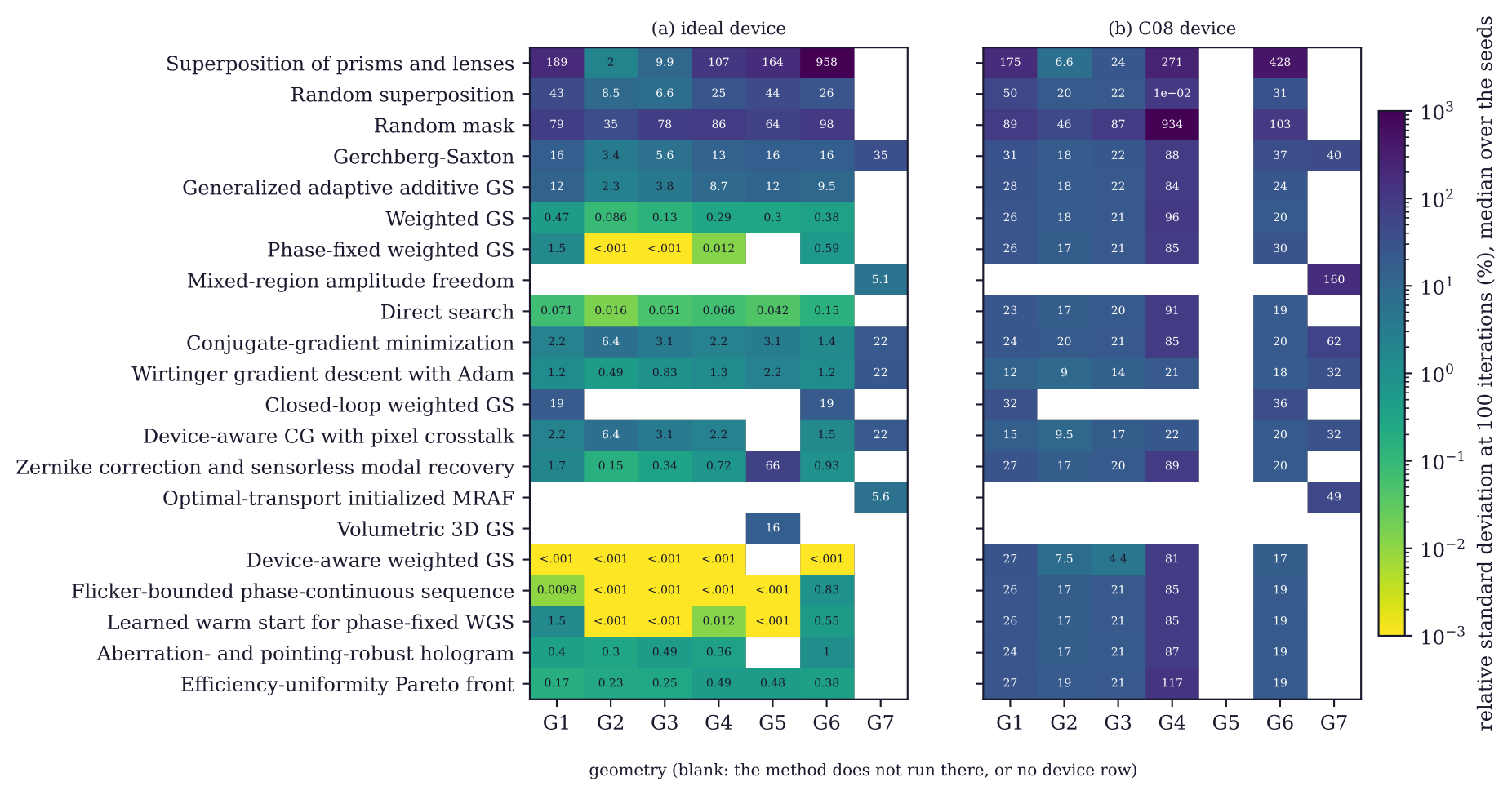
(a) ideal device
(b) C08 device
Superposition of prisms and lenses
Random superposition
Random mask
Gerchberg-Saxton
Generalized adaptive additive GS
Weighted GS
Phase-fixed weighted GS
Mixed-region amplitude freedom
Direct search
Conjugate-gradient minimization
Wirtinger gradient descent with Adam
Closed-loop weighted GS
Device-aware CG with pixel crosstalk
Zernike correction and sensorless modal recovery
Optimal-transport initialized MRAF
Volumetric 3D GS
Device-aware weighted GS
Flicker-bounded phase-continuous sequence
Learned warm start for phase-fixed WGS
Aberration- and pointing-robust hologram
Efficiency-uniformity Pareto front
189 2 9.9 107 164 958
43 8.5 6.6 25 44 26
79 35 78 86 64 98
16 3.4 5.6 13 16 16 35
12 2.3 3.8 8.7 12 9.5
0.47 0.086 0.13 0.29 0.3 0.38
1.5 <.001 <.001 0.012 0.59
5.1
0.071 0.016 0.051 0.066 0.042 0.15
2.2 6.4 3.1 2.2 3.1 1.4 22
1.2 0.49 0.83 1.3 2.2 1.2 22
19 19
2.2 6.4 3.1 2.2 1.5 22
1.7 0.15 0.34 0.72 66 0.93
5.6
16
<.001 <.001 <.001 <.001 <.001
0.0098 <.001 <.001 <.001 <.001 0.83
1.5 <.001 <.001 0.012 <.001 0.55
0.4 0.3 0.49 0.36 1
0.17 0.23 0.25 0.49 0.48 0.38
175 6.6 24 271 428
50 20 22 1e+02 31
89 46 87 934 103
31 18 22 88 37 40
28 18 22 84 24
26 18 21 96 20
26 17 21 85 30
160
23 17 20 91 19
24 20 21 85 20 62
12 9 14 21 18 32
32 36
15 9.5 17 22 20 32
27 17 20 89 20
49
27 7.5 4.4 81 17
26 17 21 85 19
26 17 21 85 19
24 17 21 87 19
27 19 21 117 19
G1 G2 G3 G4 G5 G6 G7
G1 G2 G3 G4 G5 G6 G7
geometry (blank: the method does not run there, or no device row)
relative standard deviation at 100 iterations (%), median over the seeds
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}